5634084

B.Tech. DEGREE EXAMINATION, NOVEMBER 2018.

Fourth Semester

Computer Science and Engineering

GRAPHICS AND IMAGE PROCESSING

(2013-14 Onwards Batches)

Time : Three hours | Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. List out the benefits of using space balls and trackballs.

2. List some applications for large-screen displays.

3. What command could we use to set the color of an OpenGL display window to dark gray? What command would we use to set the color of the display window to white?

4. Define the term shearing.

5. What are boundary intersection methods?

6. What is Hotelling Transformation?

7. What is the need for image Transformation?

8. What is image degradation?

9. List out the compression standards.

10. What is Morphological operation?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. List the different input and output components that are typically used with virtual-reality systems. Also, explain how users interact with a virtual scene displayed with different output devices, such as two-dimensional and stereoscopic monitors.

Or

12. Describe the existing interactive picture construction techniques.

UNIT II

13. Write a complete program to implement the Cohen-Sutherland line-clipping algorithm.

Or

14. Discuss the different line drawing algorithms with advantages and disadvantages.

UNIT III

15. Describe the various image transformation techniques.

Or

16. What is image sampling and quantization? Explain with suitable examples.

UNIT IV

17. Explain the various image restoration techniques.

Or

18. Briefly discuss the benefits of image smoothening and sharpening with advantages and disadvantages.

UNIT V

19. Discuss and elaborate on image compression algorithms.

Or

20. List out the advantages of image segmentation and also discuss its techniques.

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5634084

B.Tech. DEGREE EXAMINATION, SEPTEMBER 2020.

Fourth Semester

Computer Science and Engineering

GRAPHICS AND IMAGE PROCESSING

(2013-14 Regulations)

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Define persistence, resolution and aspect ratio.

2. What is color CRT monitor?

3. Distinguish between window and viewport.

4. What is shearing?

5. What are the applications of image processing?

6. List out the image types.

7. What is spatial filtering?

8. Define image degradation.

9. What is thresholding?

10. Define Region based segmentation.

PART B — (5 × 11 = 55 marks)

Answer ALL the questions ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Describe the various types of video display devices.

Or

12. Illustrate the interactive picture construction technique.

UNIT II

13. Describe the basic 2D dimensional transformations.

Or

14. Discuss the line drawing methods.

UNIT III

15. How to store the image and explain its file formats?

Or

16. Illustrate the Walsh and Discrete cosine transform.

UNIT IV

17. Differentiate (a) Image restoration and image enhancement (b) Periodic noise and impulse noise.

Or

18. Describe the frequency domain filtering.

UNIT V

19. Differentiate between lossy and lossless compression algorithms.

Or

20. Describe the morphological operations to extract the boundary of object for binary image.

5634084

B.Tech. DEGREE EXAMINATION, MAY 2018.

Fourth Semester

Computer Science and Engineering

GRAPHICS AND IMAGE PROCESSING

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. What is aspect ratio of a display device?
2. What is frame buffer? How it is refreshed?
3. Find the intermediate points of a line with two endpoints (0,0) and (3,3) using Bresen hamsline drawing algorithm.
4. What is window port to viewport mapping?
5. What is sampling and quantization?
6. Differentiate time domain and frequency domain representation of images.

7. What is image histogram?

8. What is salt and pepper noise and how it can be reduced?

9. What are the advantages of JPEG image compression method?

10. What are the disadvantages of thresholding edge detection methods?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

UNIT I

11. Explain in detail about various input and output devices. (11)

Or

12. Discuss in detail about interactive construction techniques. (11)

UNIT II

13. Write down the DDA line drawing algorithm and find out the intermediate points between (10,10) to (21,19) using DDA algorithm. (11)

Or

14. Write down the transformation matrix for translation, scaling, rotation, shearing and reflection. Show that two consecutive rotations are additive in nature. (11)

UNIT III

15. Explain in detail about Fourier transform and properties. (1

Or

16. Explain in detail about various image acquisitio methods. (1

UNIT IV

17. How the images can be enhanced using spatia filters? Explain in detail about various smoothin and order statistics filters. (1

Or

18. Explain in detail about histogram equalizatio process. (1

UNIT V

19. Explain in detail about basic and adaptiv thresholding segmentation algorithms. (1

Or

20. Explain in detail about transform codin technique for image compression. (1

5634084

B.Tech. DEGREE EXAMINATION.
SEPTEMBER 2021.

Fourth Semester

Computer Science Engineering

GRAPHICS AND IMAGE PROCESSING

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Why are shadow mask methods commonly used in raster scan systems including color TVs?

2. Where is the picture definition stored in memory?

3. How do you find the slope of a straight line?

4. What do you mean by composite transformations?

5. What is Pseudo color image?

6. List the different types of lighting.

7. Mention any four factors responsible for quality of an image.

8. What is a Smoothing filter?

9. What is Predictive coding?

10. What is an edge?

SECTION B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Describe the Raster - Scan systems.

Or

12. Discuss the function of the following I/O devices :

(a) Mouse

(b) Laser printer

UNIT II

13. Describe the DDA and Bresenham's line drawing algorithms.

Or

14. Explain how scaling, rotation and translation of objects are performed.

UNIT III

15. Discuss on image processing applications.

Or

16. Describe the Image Sampling and quantizat[illegible] process.

UNIT IV

17. Illustrate the Spatial filtering concepts.

Or

18. Discuss the noise categories based on distribut[illegible] and nat[illegible]e.

UNIT V

19. Discuss the significance of redundancy in ima[illegible] compression and its types.

Or

20. Explain the second order derivative filters.

5634084

B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Fourth Semester

Computer Science

GRAPHICS AND IMAGE PROCESSING

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

1. What is meant by aspect ratio?

2. List some Graphical Output devices.

3. What is meant by shearing?

4. Why are Homogeneous Coordinates used in Computer Graphics?

5. Specify the elements of DIP system.

6. List any three Image file formats.

7. What is meant by histogram equalization?

8. Give the mask used for high boost filtering.

9. What is run length coding?

10. Define Gradient Operator.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. With a neat diagram. Explain the working of a Cathode-Ray Tube.

Or

12. Write in detail about any two interactive picture construction techniques.

UNIT II

13. Explain the Cohen Sutherland Clipping algorithm.

Or

14. Explain the various transformations operations.

UNIT III

15. Explain the significance of sampling and quantization in image processing.

Or

16. Discuss the properties and applications of
(a) Hadamard transform
(b) Discrete Cosine Transforms.

UNIT IV

17. Explain image enhancement in the freque[...] domain.

Or

18. Explain image degradation model/restorat[...] process in detail.

UNIT V

19. Explain the schematics of image compress[...] standard JPEG.

Or

20. What is meant by region based segmentati[...] Explain in detail.

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19. (a) Bring out the causes of intensity changes in an image.

(b) Describe the process of edge detection with examples.

Or

20. Enumerate the process of removing imperfections in the structure of an image through morphological operations.

5634084

B.Tech DEGREE EXAMINATION, NOVEMBER 2019.

Fourth Semester

Computer Science and Engineering

GRAPHICS AND IMAGE PROCESSING

(2013–14 onwards)

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

1. List out the devices that gets data into a computer.
2. Elucidate the way in which the process of direct screen interaction is performed.
3. State how the graphics information is transformed into some useful form. Give its types.
4. Write the meaning of the term reflection.
5. List out the advantages of image processing.
6. How an image is acquired and utilized for processing?
7. Why image filtering is done in the frequency domain?

8. How can we maintain the contrast between bright and dark regions in an image?

9. List out the different types of redundancies in digital image compression.

10. Bring out anyone use of threshold in image processing.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions.

11. Elaborate on the three different types of video display devices with their working procedure and a neat structure. List out the characteristics of video display devices.

Or

12. Elucidate hard copy devices. Bring out its types and working procedure with its structure.

13. (a) What is line drawing algorithm in computer graphics?

(b) Consider the line from (0,0,) to (-8, -4), use general Bresenham's line algorithm to rasterize this line. Evaluate and tabulate all the steps involved.

Or

14. (a) Illustrate the process of rotation in computer graphics.

(b) Rotate a triangle with vertices at original coordinates (10,20), (5,10), (20,10) by 45 degrees. Plot the original and resultant triangles.

15. List out different types of image formats. Illustrate their storage format.

Or

16. (a) Describe the process of Fourier transform and explain how it works.

(b) List out the properties of Fourier transform.

17. (a) How do you make a histogram in image processing?

(b) How can we make contrast adjustment in an image using the image's histogram? Explain.

Or

18. How linear filters are used in image smoothing? Explain with examples.

5634084

B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Fourth Semester

Computer Science and Engineering

GRAPHICS AND IMAGE PROCESSING

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. What is meant by resolution?

2. What are the various Graphical input Devices?

3. What is meant by shearing?

4. What are the various line drawing methods?

5. What is the need for Image Transform?

6. What is HAAR transform?

7. What is frequency Domain filtering?

8. What is meant by image smoothening and image sharpening?

9. State the different types of redundancy.

10. Define compression ratio.

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

All questions carry equal marks.

UNIT I

11. Explain with neat diagrams the various types of video display devices.

Or

12. Explain the Interactive picture construction techniques.

UNIT II

13. Explain the polygon clipping algorithm.

Or

14. Explain the method for 2D Transformations.

UNIT III

15. Discuss in detail Image Sampling and Quantization.

Or

16. Explain Walsh Transform and Hadamard Transform.

UNIT IV

17. Discuss in detail the Image Restoration Techniques.

Or

18. Explain the Histogram Techniques for Image Enhancement.

UNIT V

19. Explain Region Based Segmentation with an example.

Or

20. Discuss in detail about morphological operations in images.

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UNIT IV

17. Describe how holomorphic filtering is used to separate illumination and reflectance component.

Or

18. How Wiener filter is helpful to reduce the mean square error when image is corrupted by motion blur and additive noise?

UNIT V

19. (a) Discuss how to construct dams using morphological operation. (6)

    (b) Describe run length encoding with example. (5)

Or

20. (a) How color image is enhanced and compare it with grayscale processing. (5)

    (b) How an image is compressed using JPEG image compression? (6)

5634084

B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2016.

Fourth Semester

Computer Science and Engineering

GRAPHICS AND IMAGE PROCESSING

(2013 – 2014 onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. How does human vision work?

2. What is sampling resolution?

3. What is Shear transformation?

4. What is mesh modeling?

5. Define and give the transfer function of Mean and Geometric mean filter.

6. Define image degradation model and sketch it.

7. Define local thresholding for edge detection.

8. Write the properties of first order and second order derivative.

9. State the need for data compression and compare lossy and lossless compression techniques

10. What are the different coloring modes?

PART B — (5 × 11 = 55 marks)

Answer ALL the questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Consider a raster system with the resolution of 1024 × 768 pixels and the color palette calls for 65,536 colors. What is the minimum amount of video RAM that the computer must have to support the above-mentioned resolution and number of colors?

Or

12. Consider two raster systems with the resolutions of 640 × 480 and 1280 × 1024.

(a) How many pixels could be accessed per second in each of these systems by a display controller that refreshes the screen at a rate of 60 frames per second? (5)

(b) What is the access time per pixel in each system? (6)

UNIT II

13. (a) Summarize midpoint circle drawing procedure.

(b) Use the above procedure to compute points on a circle with center at (5,5) and radius of 8 units.

Or

14. (a) Rotate a triangle [(4,6), (2,2) (6,2)] above the vertex (4,6) by 180 CCW and find the new vertices.

(b) Prove that Reflection is equal to rotation by 180.

UNIT III

15. Obtain DFT for the given vectros. Input image matrix = [0.0; 255 255] [2 × 2] matrix. Also analyze how the FT is used if the image is rotated or translated.

Or

16. (a) Describe how the image is digitized by sampling and quantization and explain about checker board effect and false contouring with neat sketch. (6

(b) Find DCT and its inverse for the following image data [0 255; 255 0] [2 × 2] matrix. (5

5634084

B.Tech. DEGREE EXAMINATION, MAY 2019.

Fourth Semester

Computer science and engineering

GRAPHICS AND IMAGE PROCESSING

(2013 – 14 onwards Regulations)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. List out few attributes of output primitives.

2. Differentiate oblique and orthogonal Projections.

3. Write down the principle of 2D clipping.

4. Define Spline curves.

5. Write the properties of Singular value Decomposition.

6. What is meant by Quantization?

7. List the two categories of Image Enhancement.

8. Define Histogram.

9. What are the types of discontinuity in Digital Image?

10. What is the need for compression?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, Choosing ONE from each unit.

All questions carry equal marks.

11. Write a detailed note on any three input and output devices used as Graphical User Inferface.

Or

12. Write a detailed note on Interactive Picture Construction techniques.

13. With suitable examples, explain :

(a) Rotational transformation

(b) Curve Clipping Algorithm

Or

14. Explain Cohen-Sutherland line clipping Algorithm.

15. Explain the applications of computer Graphics in the area of Image Processing.

Or

16. Explain Hadamard transformation in detail.

17. Explain the properties of 2D fourier transform.

Or

18. What are image sharpening filters? Explain the various types of sharpening filters.

19. Define thresholding and explain the various methods of thresholding in detail.

Or

20. Explain Lossless compression Algorithm in detail.

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UNIT V

19. Explain the schematics of image compression standard JPEG. (11)

Or

20. Explain Boundary descriptors in detail with a neat diagram. (11)

5634084

B.Tech. DEGREE EXAMINATION, NOVEMBER 2017.

Fourth Semester

Computer Science and Engineering

GRAPHICS AND IMAGE PROCESSING

(2013–14 onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. What is bitmap and what is pixmap?

2. Draw the spectral response function curve of three types of cone on human retina.

3. What do you mean by scan conversion?

4. Define Affine transformation.

5. Define weber ratio.

6. What are the elements of visual perception.

7. Compare spatial and frequency Domain methods.

8. What is thresholding? Explain.

9. Define Huffman coding.

10. Define Gradient Operator?

PART B — (5 × 11 = 55 marks)

Answer ALL the questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain and differentiate the functionality of LED and LCD. (11)

Or

12. Explain Rubber band method, Zooming, Panning and Dragging. (11)

UNIT II

13. Explain Bresenhams line drawing algorithm. Digitize a line from (20,10) to (30,18) on a raster screen using bresenham's line drawing algorithm. (11)

Or

14. Find the Clipping coordinates for the line p1p2 where p1=(10,10) and (11) p2(60,30), against the window with (Xmin,Ymin) = (15,15) and (Xmax,Vmax) = (25,25). Use Liang Barsky algorithm. (11)

UNIT III

15. Write a detailed note on Walsh transform. (11)

Or

16. Discuss the properties and applications of

(a) Hadamard transform

(b) Discrete Cosine Transforms (11

UNIT IV

17. Explain Homomorphic filtering in detail. (11

Or

18. What are the two approaches for blind imag restoration? Explain in detail. (11